



COURSE SYLLABUS

UNIT I — INTRODUCTION TO COMPUTER NETWORKS AND PHYSICAL LAYER

Subject: Computer Network

Branch: Computer Engineering | Sem: V | SPPU 2024 Pattern

SYLLABUS TOPICS:

- Introduction to computer networks
- Uses of computer networks - business applications, home applications, mobile users
- Network hardware - PAN, LAN, MAN, WAN and internetworks
- Network software - protocol hierarchies, design issues for layers, connection-oriented and connection-less services, service primitives, relationship between services and protocols
- Reference models - OSI and TCP/IP models
- Physical Layer: guided transmission media
- Wireless transmission
- Telephone system
- Narrowband and broadband communication systems

1.1 Introduction to Computer Networks

Definition:

A computer network is an interconnected collection of autonomous computers and other devices that are able to exchange information and share resources using a common set of rules called protocols.

Introduction / Need:

Before networks existed, every computer worked as an isolated island. Data had to be copied on to a disk and carried physically to another machine, a method jokingly called "sneakernet" (moving data by walking, wearing sneakers, instead of using a wire).

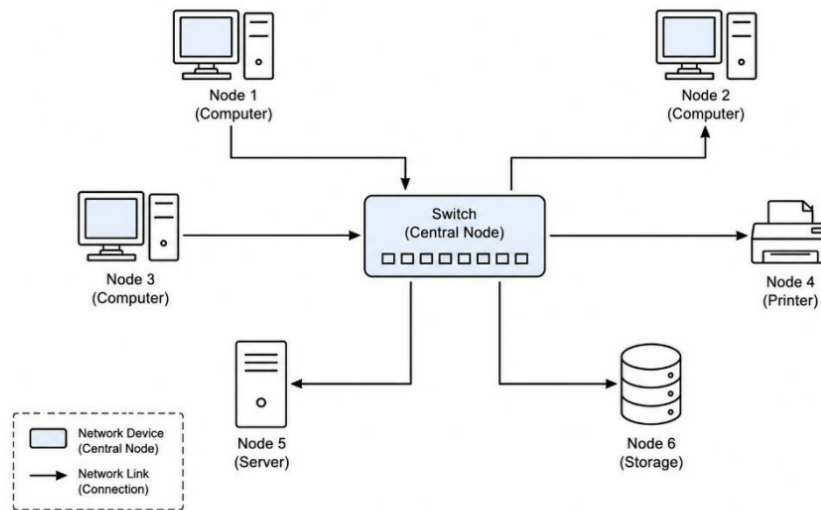
Networks were built to remove this physical dependency. They let multiple computers talk directly over a shared communication path, so information, printers, and processing power can be shared instantly instead of manually.

1.1.1 What a Network Really Is

Two computers are said to be interconnected if they are able to exchange information. The connection does not have to be a copper wire; it can be fibre optics, radio waves, microwave links, or satellites.

The word autonomous is important here. Each computer in the network can boot, run, and shut down on its own. A network is different from a multiprocessor system, where one master processor commands the rest.

Autonomy also means every machine keeps its own identity, operating system, and storage, while still being able to cooperate through the network.

Fig 1.1 — Basic Computer Network showing interconnected nodes**KEY POINTS — Introduction to Computer Networks**

- A network needs two things: connected autonomous devices and a common protocol to talk in.
- Do not confuse network with distributed system — a distributed system hides the presence of multiple computers from the user, while in a network the user is aware of separate machines.
- Common exam trap: students write "network is group of computers connected by cable" — always add autonomy and protocol in the definition to score full marks.
- SPPU frequently asks to differentiate network vs distributed system in this topic.
- Remember: sneakernet is a term used to describe manual data transfer before networking existed.

SPPU EXPECTED QUESTIONS — Introduction to Computer Networks

Q1. Define computer network. Explain the concept of autonomous systems in a network. (5 marks)

Q2. Differentiate between a computer network and a distributed system with examples. (6 marks)

1.2 Uses of Computer Networks

Definition:

The uses of computer networks describe the practical purposes for which organisations, homes, and individual mobile users deploy networking technology, ranging from resource sharing to communication and entertainment.

Introduction / Need:

Networks are not built for their own sake; they solve real problems for three broad groups of users — businesses, home users, and mobile users. Each group has a different motivation for going online.

Studying these uses helps an engineer decide what kind of network to design. A business network prioritises reliability and security, while a home network prioritises cost and ease of use.

1.2.1 Business Applications

Companies use networks mainly to share expensive resources such as printers, scanners, and storage servers among many employees, instead of buying one device per desk.

Resource sharing (using one physical resource by many users, so it does not sit idle) reduces cost quickly in large offices.

Networks also allow information sharing through client-server computing. A server holds the data (for example, a database of customer orders), and client machines request and update this data over the network.

Businesses use networks for electronic commerce, business-to-business (B2B) transactions, video conferencing between branch offices, and Voice over IP (VoIP) telephony to cut communication cost.

Sr. No.	Business Use	Purpose
1	Resource Sharing	Multiple employees use one printer or server, saving hardware cost
2	Client-Server Model	Central server stores data; clients request and update records
3	E-mail and Messaging	Fast internal and external communication replacing paper memos
4	E-commerce	Selling products and services online (B2B and B2C transactions)
5	Video Conferencing	Remote meetings between branch offices, saving travel cost

1.2.2 Home Applications

Home users connect to networks mainly for access to remote information, person-to-person communication, entertainment, and increasingly, for the Internet of Things (IoT).

Access to remote information includes browsing the web, online banking, and reading digital newspapers. Person-to-person communication covers e-mail, video calls, and social networking.

Entertainment use covers streaming video and music, online gaming, and downloading content. IoT use covers smart home devices such as thermostats, cameras, and voice assistants that connect to the home network.

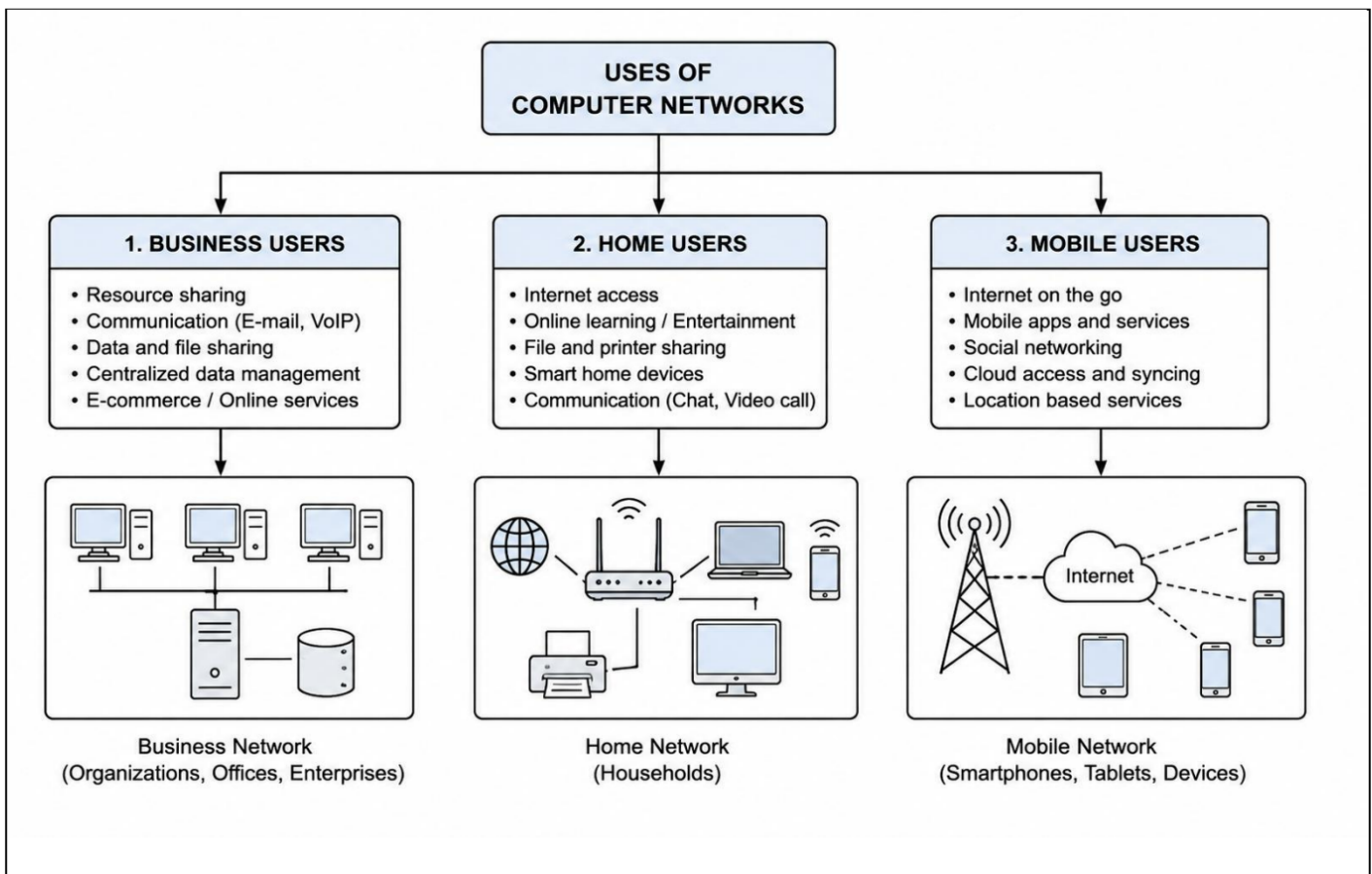
1.2.3 Mobile Users

Mobile users need access to information regardless of their physical location, using laptops, smartphones, and tablets connected through cellular or Wi-Fi networks.

Typical mobile network uses include checking e-mail while travelling, using GPS-based navigation, mobile banking, and location-based services that push offers based on the user's current position.

Mobile commerce (m-commerce) allows payments and purchases directly from a handheld device, which has become extremely common with UPI and digital wallet systems.

Fig 1.2 — Classification of network uses - business, home, and mobile users



Worked Example:

Given: A company wants employees at 3 branch offices to access one central customer database and hold weekly video meetings.

Find: Which category of network use fits, and what services are required.

Solution:

Step 1: Identify the requirement type: shared central data and real-time communication.

Step 2: Map data sharing requirement to Business Application - Client-Server model.

Step 3: Map video meeting requirement to Business Application - Video Conferencing.

Step 4: Conclude that both uses fall under Business Applications of computer networks.

Ans: The scenario represents **Business Applications**, specifically **client-server resource sharing and video conferencing**.

Applications:

- Online banking portals used by home users for fund transfer and bill payment
- Corporate ERP systems (e.g., SAP) used by businesses for centralised data management
- Google Maps and Uber using mobile network location services for navigation and ride booking
- Netflix and Spotify using home network bandwidth for streaming entertainment

KEY POINTS — Uses of Computer Networks

Three categories always asked together: Business, Home, Mobile — never mix their examples while answering.

Resource sharing and client-server computing are the two pillars of business use.

Common exam trap: students forget IoT under home applications — always mention smart devices.

M-commerce is the most confused term with e-commerce — m-commerce is strictly through mobile/handheld devices.

SPPU often asks for real-life examples for each category — keep at least one ready for each.

SPPU EXPECTED QUESTIONS — Uses of Computer Networks

Q1. Explain business applications of computer networks with suitable examples. (5 marks)

Q2. Describe home applications and mobile user applications of computer networks. (6 marks)

Q3. Compare business, home, and mobile uses of computer networks. (6 marks)

1.3 Network Hardware — PAN, LAN, MAN, WAN and Internetworks

Definition:

Network hardware classifies computer networks by their physical scale and transmission technology, ranging from a Personal Area Network (PAN) covering a few metres to a Wide Area Network (WAN) that spans countries.

Introduction / Need:

Not every network problem needs the same solution. Connecting a mouse to a laptop is very different from connecting Mumbai and Pune offices of a bank.

Classifying networks by physical distance (scale) helps engineers pick the right transmission media, speed, and protocol for the situation, instead of using one technology everywhere.

1.3.1 Personal Area Network (PAN)

A PAN connects devices within the reach of a single person, typically within 1 to 10 metres, such as a wireless keyboard, mouse, headset, and smartwatch talking to a laptop or phone.

PANs commonly use Bluetooth or Near Field Communication (NFC) instead of wires, which removes cable clutter around a work desk.

PAN devices form a master-slave setup: the mobile phone or laptop acts as master, and accessories act as slaves that only respond when addressed.

1.3.2 Local Area Network (LAN)

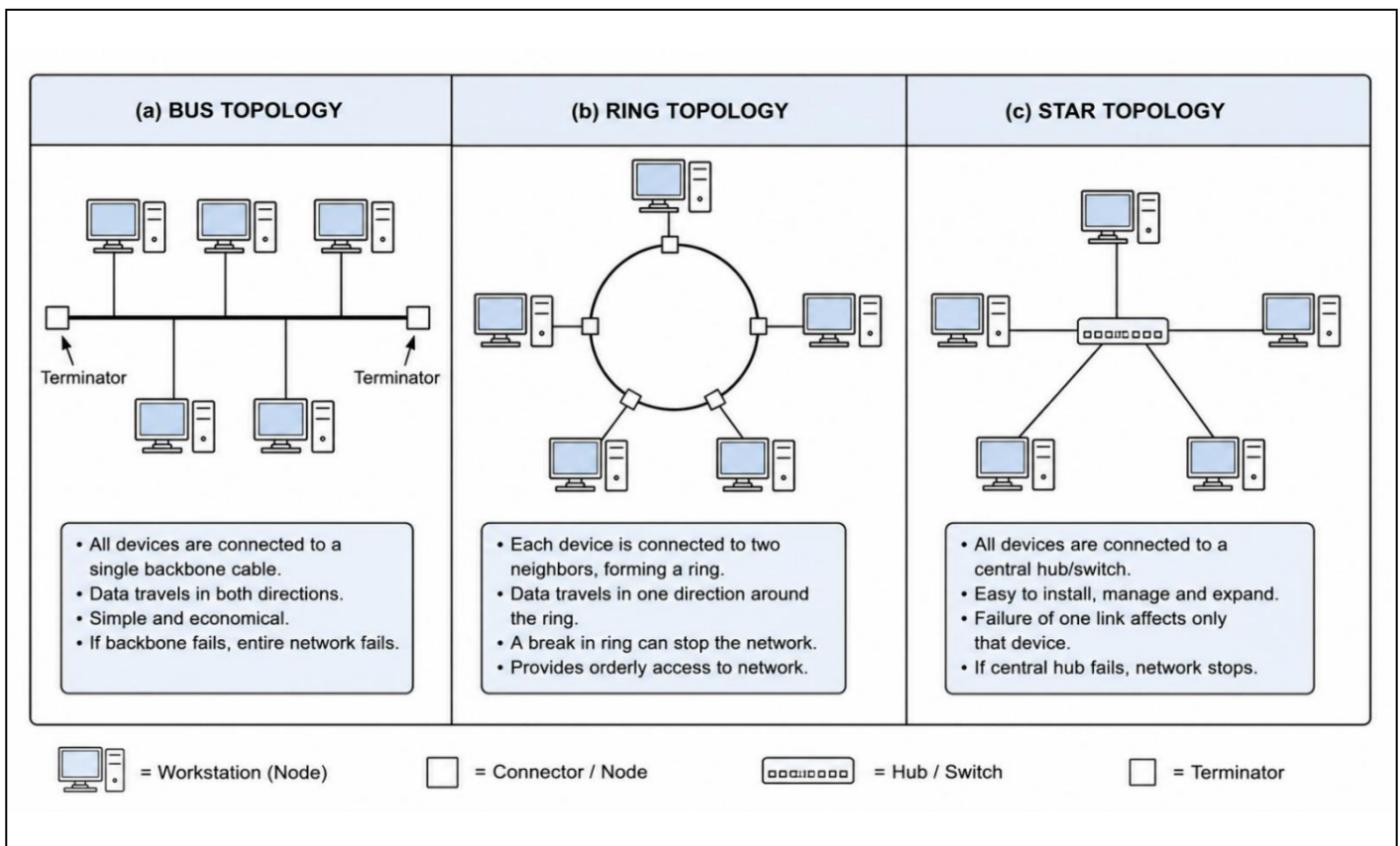
A LAN is a privately owned network that operates within a single building or campus, typically up to a few kilometres, connecting personal computers and devices so they can share resources and exchange information.

LANs are distinguished by three main factors: size (restricted, so worst-case transmission time is bounded), transmission technology (usually a single cable that all machines share), and topology (bus, ring, or star layout).

Traditional LANs run at 10 Mbps to 10 Gbps speeds, have low delay, and make very few errors because the wiring is fully owned and controlled by the organisation.

Two broad technologies dominate LANs: broadcast networks such as Ethernet, where every machine sees every frame, and switched networks, where a switch forwards frames only to the intended destination port.

Fig 1.3 — LAN in bus, ring, and star topology



1.3.3 Metropolitan Area Network (MAN)

A MAN covers a whole city, typically up to 50 km, connecting several LANs together — for example, linking all branch offices of a bank spread across a city through a shared cable TV or fibre backbone.

A well-known MAN example is cable television network reused for Internet access, or a city-wide fibre ring built by a telecom operator.

MANs are owned either by a single company or by a consortium of users, unlike LANs which are always privately owned by one organisation.

1.3.4 Wide Area Network (WAN)

A WAN spans a large geographical area, often a country or continent, and connects multiple LANs and MANs using long-distance transmission media such as leased telephone lines, satellite links, or undersea fibre cables.

A WAN contains two distinct components: hosts (end-user machines running application programs) and the communication subnet (the collection of routers and transmission lines that carry messages between hosts).

In a subnet, routers connect two or more transmission lines. When data arrives on an incoming line, the router must choose an outgoing line to forward it on, using a technique called store-and-forward, or packet-switched, operation.

The design of the subnet can follow a point-to-point layout (dedicated links between router pairs, as in most WANs) or a broadcast layout (shared channel, as in satellite-based WANs).

1.3.5 Internetworks

An internetwork is formed when two or more distinct networks — possibly using different underlying technologies — are connected together, most commonly through a device called a gateway or router.

The Internet itself is the largest example of an internetwork: it interconnects millions of LANs, MANs, and WANs owned by different organisations across the world using the common TCP/IP protocol suite.

The key benefit of internetworking is that a user on one type of network (say, an office LAN) can communicate transparently with a user on a completely different network (say, a home broadband WAN) without knowing the internal technology differences.

Table: Comparison of PAN, LAN, MAN, and WAN

Sr. No.	Parameter	PAN	LAN	MAN / WAN
1	Coverage Area	1 - 10 metres	Up to a few km (building/campus)	MAN: up to 50 km, WAN: country/continent
2	Ownership	Personal device	Single organisation	MAN: consortium, WAN: multiple ISPs
3	Typical Speed	1 - 3 Mbps (Bluetooth)	10 Mbps - 10 Gbps	Varies, generally lower than LAN
4	Error Rate	Low (short range)	Very low (private wiring)	Higher due to long distance links
5	Example Technology	Bluetooth, NFC	Ethernet, Wi-Fi	Leased lines, satellite, fibre backbone
6	Example	Wireless earphones with phone	Office LAN with switch	Internet, bank city-wide network

Worked Example:

Given: A bank has 20 computers in one branch building, all its city branches need to be linked, and its branches across India also need to communicate.

Find: Identify which network category applies to each requirement.

Solution:

Step 1: 20 computers inside one building fall under a single, privately owned, small-area network.

Step 2: This matches the LAN definition exactly, since it is within one building.

Step 3: Connecting all branches within one city matches the MAN definition, since the area is limited to a city.

Step 4: Connecting branches across India, using leased lines or satellite links, matches the WAN definition.

Ans: Branch-level network is a LAN, city-level linking is a MAN, and country-level linking is a WAN.

Advantages of Classifying Networks by Scale:

- Correct technology choice — engineers pick appropriate media and speed for each scale
- Cost control — small-scale PAN/LAN avoid the expense of WAN-grade equipment where not needed
- Better fault isolation — a LAN fault does not automatically bring down the entire WAN
- Simplifies planning — administrators can budget and expand network in clear stages
- Enables internetworking — clear scale boundaries make it easy to design gateways between them

Disadvantages / Limitations:

- Rigid classification does not always match real deployments (e.g., large campus LANs behave like small MANs)
- MAN technology has become less distinct as fibre WAN and LAN speeds have converged
- Scaling from LAN to WAN often needs a complete redesign of addressing and routing
- Multiple ownership in MAN/WAN increases coordination and security complexity

Applications:

- Bluetooth PAN used in wireless earbuds such as Apple AirPods pairing with a phone
- Ethernet LAN used inside SPPU college computer laboratories for resource sharing
- Cable-TV based MAN used by city broadband providers like Hathway for last-mile Internet
- Leased-line WAN used by banks such as SBI to connect branches across the country
- The global Internet, the largest internetwork, connecting ISPs, universities and enterprises worldwide

KEY POINTS — Network Hardware

Scale order to remember: PAN < LAN < MAN < WAN < Internetwork (network of networks).

A subnet in WAN context means routers + transmission lines, not to be confused with IP subnetting.

Common exam trap: students say MAN and WAN are the same — MAN is city-limited, WAN is unlimited geography.

Most confused comparison: LAN vs MAN vs WAN table is asked almost every year — memorise the comparison table.

Internet is always the go-to example of an internetwork in SPPU papers.

SPPU EXPECTED QUESTIONS — Network Hardware

Q1. Explain the classification of networks based on scale with examples. (6 marks)

Q2. Differentiate between LAN, MAN and WAN with a comparison table. (8 marks)

Q3. Explain the structure of a WAN subnet with a neat diagram. (6 marks)

Q4. What is an internetwork? Explain with the example of the Internet. (5 marks)

Q5. Describe PAN with its typical technology and applications. (5 marks)

1.4 Network Software

Definition:

Network software refers to the organised set of protocols, layered structures, and rules that govern how data is formatted, addressed, transmitted, and received across a computer network.

Introduction / Need:

Early network designers realised that writing one giant program to handle everything — wiring, addressing, error control, and applications — was unmanageable and error-prone.

The solution was to break the problem into layers, each handling one specific job and offering a clean, simple interface (a well-defined set of operations offered to the layer above it) to the layer above it. This idea is called protocol hierarchy.

1.4.1 Protocol Hierarchies

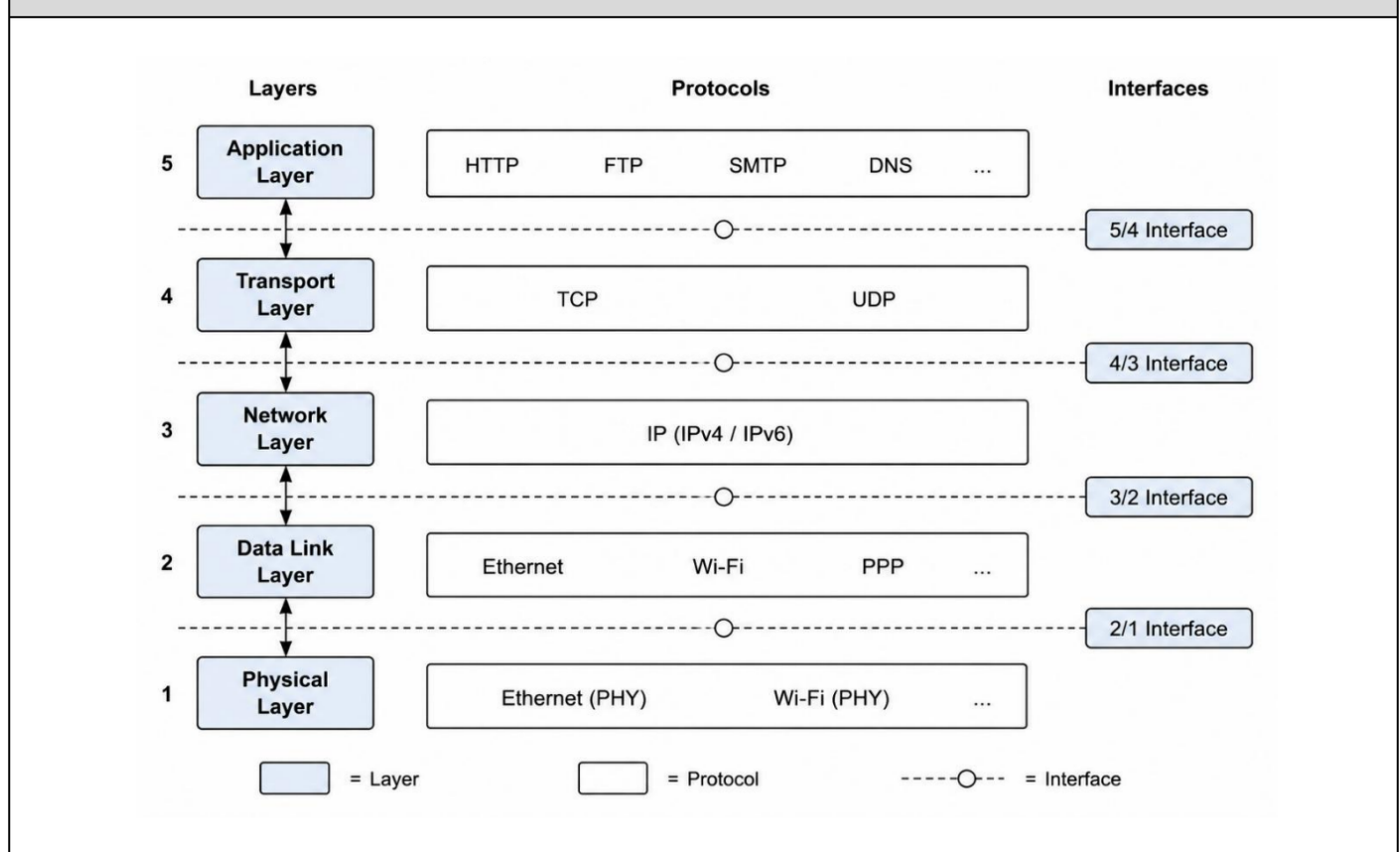
A protocol is an agreement between the communicating parties on how communication is to proceed — like a common language and set of rules both sides must follow.

Layer n on one machine logically talks to layer n on another machine using the rules of the layer n protocol. In reality no data is transferred directly between peer layers; each layer passes data down to the layer below it until it reaches the physical medium.

Between each pair of adjacent layers there is an interface, which defines what primitive operations and services the lower layer offers to the upper layer.

A set of layers and protocols together is called a network architecture. A list of protocols used by a system, one protocol per layer, is called a protocol stack.

Fig 1.5 — Layers, protocols, and interfaces in a protocol hierarchy



1.4.2 Design Issues for the Layers

Several fundamental issues repeat at almost every layer of a network architecture, and SPPU frequently asks students to list them.

Every layer needs a mechanism for addressing, so that a process on one machine can specify exactly which process on another machine it wants to talk to.

Sr. No.	Design Issue	Explanation
1	Addressing	Mechanism to identify sender and receiver process uniquely on the network

Sr. No.	Design Issue	Explanation
2	Error Control	Agreeing on error-detecting and error-correcting codes so both ends know when data is damaged
3	Flow Control	Mechanism to stop a fast sender from overwhelming a slow receiver's buffer
4	Multiplexing / Splitting	Deciding whether to combine several conversations onto one channel, or split one conversation across several
5	Routing	Determining the correct path when multiple routes exist between source and destination
6	Message Sequencing	Ensuring messages that arrive out of order are reassembled in correct order at the receiver

1.4.3 Connection-Oriented and Connectionless Services

A connection-oriented service works like the telephone system: the sender and receiver first establish a connection (a virtual path), use it to send data in order, and then release it — much like dialling, talking, and hanging up.

A connectionless service works like the postal system: each message (or packet) carries the full destination address and is routed independently, with no guarantee that packets arrive in the order they were sent.

Connection-oriented service is typically reliable, since data is acknowledged, sequenced, and flow-controlled. Connectionless service is typically unreliable (also called best-effort), since no acknowledgement or ordering guarantee is made.

Some connectionless services provide acknowledgement without a formal connection, called acknowledged datagram service, used where individual message delivery must be confirmed but a full connection is unnecessary overhead.

Sr. No.	Parameter	Connection-Oriented Service	Connectionless Service
1	Setup	Requires connection establishment before data transfer	No setup; data sent directly
2	Analogy	Telephone system	Postal / letter system
3	Packet Ordering	Guaranteed in-order delivery	Packets may arrive out of order
4	Reliability	Reliable, generally acknowledged	Best-effort, mostly unacknowledged
5	Example Protocol	TCP	UDP, IP

1.4.4 Service Primitives

A service is formally specified by a set of primitives (operations) that are available to a user process to access that service, such as request, indication, response, and confirm.

The four basic primitives work as follows: a request primitive asks the service to do some work, an indication primitive informs a peer entity that an event has happened, a response primitive is used by a peer entity to reply to an earlier indication, and a confirm primitive carries the results of an earlier request back to the caller.

These primitives are used in a defined sequence during connection establishment — for example, a server issues LISTEN, the client issues CONNECT (request), the server receives CONNECTION INDICATION, the server issues ACCEPT (response), and finally the client receives CONNECT CONFIRMED.

```
// Simplified service primitive sequence for connection setup
Server: LISTEN()           -> waits passively for a connection
Client: CONNECT(dest_addr) -> sends a CONNECTION REQUEST packet
Server: <-- CONNECTION INDICATION received
Server: ACCEPT()           -> sends a response back to client
Client: <-- CONNECT CONFIRMED received, connection established
Both:   SEND(data) / RECEIVE(data) -> exchange data over connection
Either: DISCONNECT()       -> releases the connection
```

Line-by-line explanation: LISTEN is issued by the server first so it is ready to accept incoming requests. CONNECT is issued by the client, carrying the destination address, and triggers a CONNECTION INDICATION at the server. ACCEPT lets the server agree to the connection, generating a CONNECT CONFIRMED at the client. Once both sides confirm, SEND and RECEIVE move the actual data, and DISCONNECT is issued by either side to close the session cleanly.

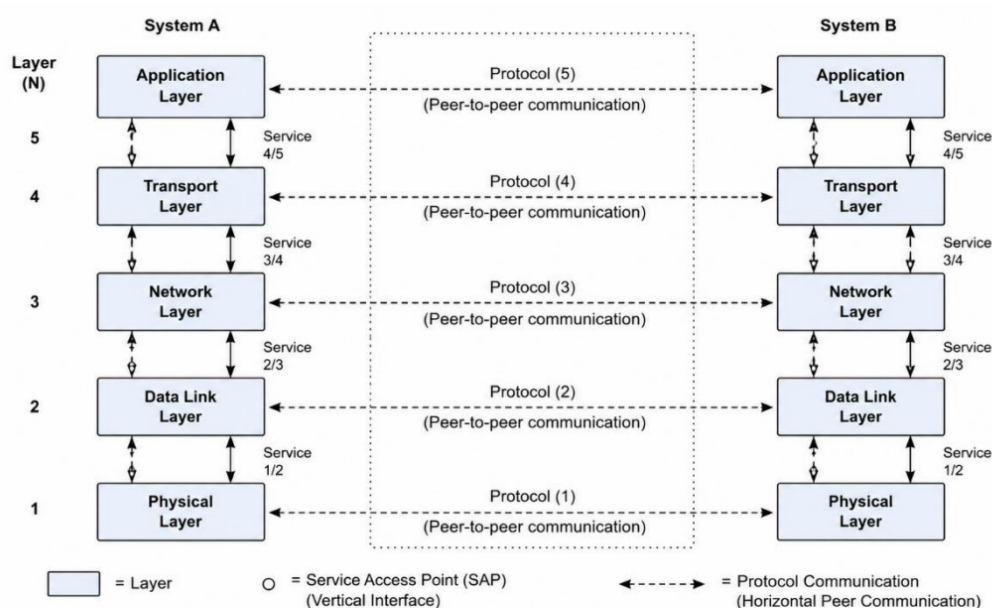
1.4.5 Relationship Between Services and Protocols

A service is a set of operations that a layer provides to the layer above it; it defines what operations the layer can perform, but says nothing about how those operations are implemented.

A protocol is a set of rules governing the format and meaning of the frames, packets, or messages exchanged by peer entities within a layer; it defines how the service is actually implemented.

A useful analogy: service is like a function's specification (what it does), and protocol is like the function's internal code (how it does it). This separation lets a layer's protocol be changed completely without affecting the layer above, as long as the service it offers stays the same.

Fig 1.6 — Relationship between service (vertical interface) and protocol (horizontal peer communication)



Worked Example:

Given: A network layer offers a reliable, ordered byte-stream delivery service to the transport layer above it.

Find: Identify whether this describes a service or a protocol, and justify.

Solution:

Step 1: Check what is being described: capability offered to the layer above (ordered byte-stream delivery).

Step 2: Recall that a service defines what operations a layer offers upward, without detailing implementation.

Step 3: Note no frame format or rule between peer entities is mentioned here, only the offered capability.

Step 4: Conclude this description matches the definition of a service, not a protocol.

Ans: This describes a service, since it defines the capability offered to the upper layer without specifying peer-to-peer rules.

Advantages of Layered Network Software:

- Modularity — each layer can be designed, tested, and upgraded independently
- Interoperability — different vendors can build compatible layer implementations
- Simplifies troubleshooting — problems can be isolated to a specific layer
- Reusability — a lower layer's service can support multiple different upper-layer protocols
- Encourages standardisation, allowing global adoption (as seen with TCP/IP)

Disadvantages / Limitations:

- Layering adds processing overhead at every layer boundary, increasing latency slightly
- Strict layering can sometimes force inefficient designs when a shortcut across layers would perform better
- Debugging across many layers can still be complex despite modularity
- Excess header information is added at each layer, increasing overall packet size

Applications:

- TCP protocol stack used by every web browser to reliably load websites
- UDP connectionless service used by video calling apps like Zoom for low-latency streaming
- Service primitive model implemented internally in socket programming (connect, listen, accept, send)
- Layered protocol hierarchy forms the basis of both OSI and TCP/IP reference models
- Flow control design issue applied in USB and network card driver buffering

KEY POINTS — Network Software

Service = what a layer offers (vertical, between layers); Protocol = how peers talk (horizontal, same layer).

Four primitives to memorise: Request, Indication, Response, Confirm.

Common exam trap: students confuse connectionless with unreliable always being true — acknowledged datagram service is an exception.

Most confused comparison: connection-oriented (TCP, telephone) vs connectionless (UDP, postal) — a favourite SPPU question.

Design issues (addressing, error control, flow control, routing) repeat at multiple layers — a favourite list-based question.

SPPU EXPECTED QUESTIONS — Network Software

- Q1. Explain the concept of protocol hierarchy with a neat diagram. (6 marks)
- Q2. What are the design issues for layers in a network architecture? (6 marks)
- Q3. Differentiate between connection-oriented and connectionless services with examples. (8 marks)
- Q4. Explain service primitives with a suitable example of connection establishment. (6 marks)
- Q5. Describe the relationship between services and protocols in layered architecture. (5 marks)

1.5 Reference Models — OSI and TCP/IP

Definition:

A reference model is a conceptual, layered framework that standardises the functions a communication system must perform, without dictating the exact protocol used at each layer, so that different networks can interoperate.

Introduction / Need:

As different vendors began building networking hardware and software independently, there was no common structure to make their products talk to each other.

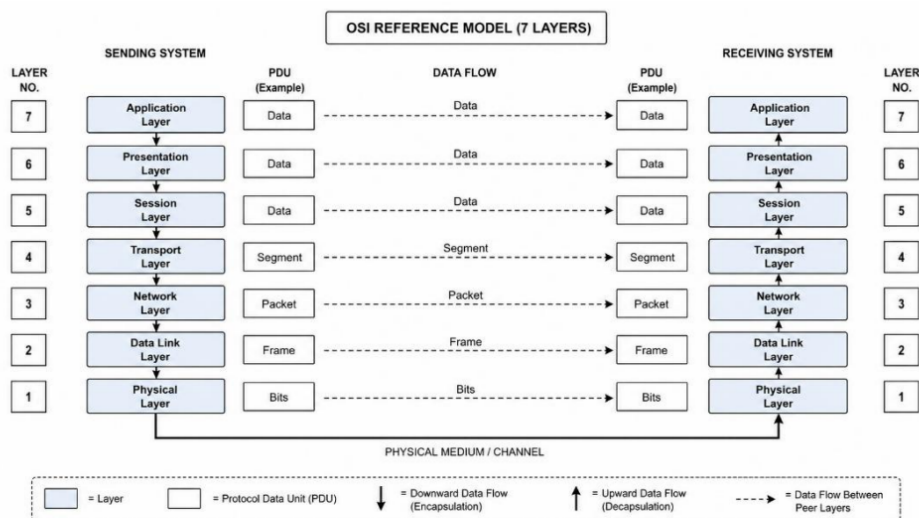
Reference models were created to give the whole industry a shared blueprint. OSI came from an international standards effort, while TCP/IP came from real, working ARPANET protocols, and today the Internet runs on TCP/IP while OSI is mainly used as a teaching model.

1.5.1 OSI Reference Model

The Open Systems Interconnection (OSI) model, developed by ISO, organises network communication into exactly seven layers, each with a clearly defined job, from the physical wire up to the user application.

The model was designed around three key principles: create a layer wherever a distinct level of abstraction is needed, keep each layer's function well defined, and choose layer boundaries to minimise the information flow across interfaces.

Sr. No.	Layer	Function
1	Physical Layer	Transmits raw bits over a physical medium; deals with voltage, cabling, and connectors
2	Data Link Layer	Organises bits into frames, performs error detection and correction, and manages access to shared media
3	Network Layer	Handles logical addressing and routing of packets across multiple interconnected networks
4	Transport Layer	Provides end-to-end reliable or unreliable delivery, segmentation, and flow control between hosts
5	Session Layer	Establishes, manages, and terminates sessions (dialogues) between two communicating applications
6	Presentation Layer	Handles data representation, encryption, and compression so both ends understand the same data format
7	Application Layer	Provides network services directly to user applications, such as file transfer, e-mail, and web browsing

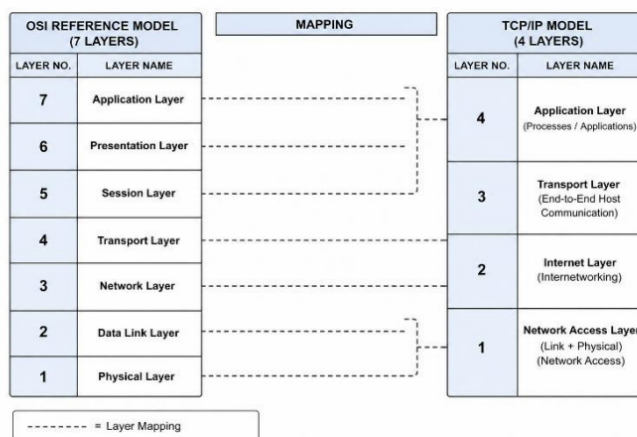
Fig 1.7 — OSI seven-layer reference model with data flow

1.5.2 TCP/IP Reference Model

The TCP/IP reference model grew out of the practical ARPANET research project and is organised into four layers, focused on real internetworking rather than theoretical completeness.

Its layers are: Link layer (handling physical transmission, roughly matching OSI's physical and data link layers), Internet layer (handling logical addressing and routing, matching OSI's network layer, based on the IP protocol), Transport layer (providing end-to-end communication, matching OSI's transport layer, using TCP or UDP), and Application layer (combining OSI's session, presentation, and application layers into one, since these distinctions were found unnecessary in practice).

The Internet layer was deliberately designed as connectionless, similar to a datagram service, so packets could be routed independently even if some network links failed — a critical requirement for the military-origin ARPANET.

Fig 1.8 — TCP/IP four-layer model mapped against OSI seven layers

1.5.3 Comparison of OSI and TCP/IP Models

Although both models use layering, they differ in philosophy, number of layers, and how strictly they separate service, interface, and protocol.

OSI clearly distinguishes between service (what a layer does), interface (how to access it), and protocol (how peers communicate) at every layer. TCP/IP did not originally make this distinction as clearly, since it grew from working code rather than a formal design process.

Sr. No.	Parameter	OSI Model	TCP/IP Model
1	Number of Layers	Seven layers	Four layers
2	Origin	ISO standards committee, designed before protocols existed	ARPANET project, protocols existed before the model
3	Service/Interface/Protocol	Clearly separated at every layer	Not clearly separated, especially in early design
4	Connection Support	Supports both connection-oriented and connectionless at network layer	Internet layer is connectionless only (IP)
5	Usage Today	Used mainly as a teaching and reference model	Used as the actual working model of the Internet
6	Layer combining Session/Presentation	Separate Session and Presentation layers exist	Merged into the Application layer

Worked Example:

Given: An application needs encryption of data before sending, and also needs reliable end-to-end delivery between two hosts.

Find: Identify which OSI layers handle each requirement, and how TCP/IP handles the same functions.

Solution:

Step 1: Encryption and data representation is the job of the OSI Presentation layer.

Step 2: Reliable end-to-end delivery is the job of the OSI Transport layer.

Step 3: In TCP/IP, Presentation-layer function is merged into the Application layer.

Step 4: In TCP/IP, Transport-layer function is directly handled by TCP.

Ans: OSI: Presentation layer for encryption, Transport layer for reliability. TCP/IP: Application layer for encryption, TCP for reliability.

Advantages of Reference Models:

- Provide a common vocabulary for engineers worldwide to discuss networking concepts
- Allow independent vendors to build interoperable hardware and software
- Simplify teaching and troubleshooting by isolating functions to specific layers
- OSI's strict layering encourages clean, reusable protocol design
- TCP/IP's practical design has proven scalable enough to run the entire global Internet

Disadvantages / Limitations:

- OSI model was never widely implemented commercially due to its complexity and late timing
- TCP/IP model lacks a formal distinction between service, interface, and protocol
- TCP/IP's four-layer structure is considered too general for describing non-IP networks
- Reference models cannot guarantee actual protocol performance; they only describe structure

Applications:

- OSI model used as the standard teaching framework in networking courses worldwide, including SPPU syllabus
- TCP/IP model is the actual protocol suite running the global Internet today
- Network troubleshooting tools like Wireshark display captured packets according to OSI/TCP-IP layers
- Router and switch manufacturers such as Cisco design certification exams around these layer models
- Firewall rules and security appliances are configured layer-wise, based on these reference models

KEY POINTS — Reference Models

OSI has 7 layers, TCP/IP has 4 layers — never mix layer counts while answering.

Mnemonic for OSI (bottom to top): Please Do Not Throw Sausage Pizza Away (Physical, Data Link, Network, Transport, Session, Presentation, Application).

Common exam trap: students think OSI is used in the Internet — actually TCP/IP runs the real Internet.

Most confused comparison: OSI vs TCP/IP layer mapping is asked almost every year — keep the mapping table ready.

TCP/IP's Internet layer is strictly connectionless (IP is a datagram protocol) — a frequently tested fact.

SPPU EXPECTED QUESTIONS — Reference Models

Q1. Explain the OSI reference model with functions of each layer. (8 marks)

Q2. Explain the TCP/IP reference model with a neat diagram. (6 marks)

Q3. Compare OSI and TCP/IP reference models. (8 marks)

Q4. Explain the design principles used while creating the OSI model. (5 marks)

Q5. Why is the Internet layer of TCP/IP model kept connectionless? Justify with reasoning. (6 marks)

1.6 Physical Layer: Guided Transmission Media

Definition:

Guided transmission media are physical, bounded pathways such as wires and cables that carry signals from one point to another along a fixed conductor or fibre, unlike wireless media which radiate signals through open space.

Introduction / Need:

The physical layer is responsible for transmitting raw bits over a communication channel, and the choice of medium directly decides how fast, how far, and how reliably those bits can travel.

Guided media are studied first because most LANs, telephone lines, and backbone links still rely on physical cables, and each cable type trades off cost, bandwidth, and noise immunity differently.

1.6.1 Magnetic Media (Sneakernet)

One of the most cost-effective ways to move large volumes of data is to physically write it to a storage medium, such as a hard disk or high-capacity tape, and courier it to the destination.

Although it sounds old-fashioned, magnetic media transport can beat network transmission on pure cost-per-bit for bulk data transfer over long distances, even though the latency (delay before data becomes usable) is very high.

1.6.2 Twisted Pair Cable

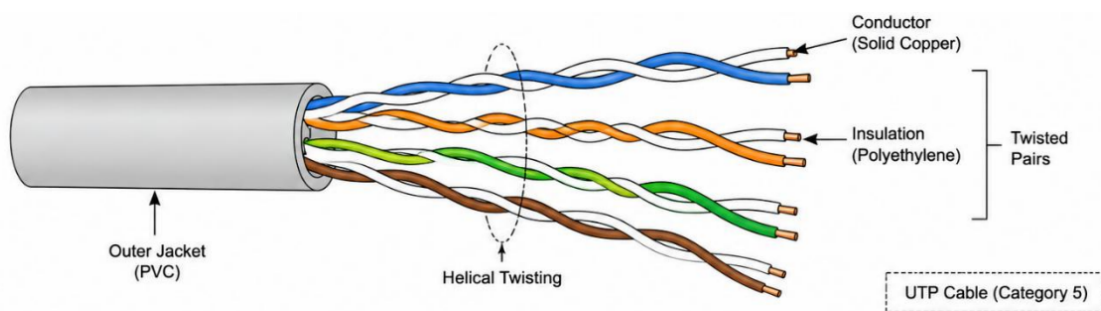
A twisted pair consists of two insulated copper wires, typically about 1 mm thick, twisted together in a helical form to reduce electrical interference from nearby pairs and external noise.

Twisting cancels out interference because the two wires pick up nearly identical amounts of unwanted signal (called crosstalk) from neighbouring pairs, which is then cancelled out at the receiver.

Twisted pair is classified as Category 3 (used in older telephone wiring, supporting lower speeds) and Category 5 or higher (used in modern LANs, supporting up to 1000 Mbps or more with tighter twists per centimetre).

Twisted pairs can carry either analog or digital signals, and their bandwidth (the range of frequencies a medium can carry) depends on wire thickness and cable distance.

Fig 1.9 — Twisted pair cable construction showing helical twisting



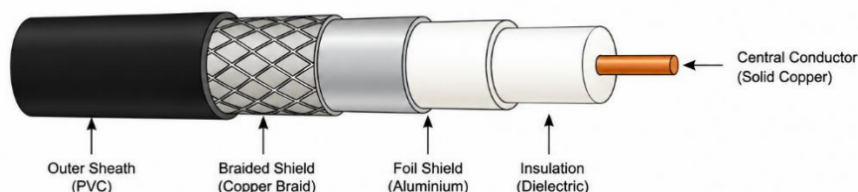
1.6.3 Coaxial Cable

Coaxial cable has a stiff copper core as the central conductor, surrounded by insulating material, a cylindrical braided outer conductor, and a protective plastic sheath.

Its construction and shielding give coaxial cable much better noise immunity, wider bandwidth, and longer usable distance compared to unshielded twisted pair.

Two types exist: 50-ohm coaxial cable, commonly used for digital transmission, and 75-ohm coaxial cable, commonly used for analog transmission such as cable television.

Fig 1.10 — Coaxial cable cross-section showing core, insulation, braid, and sheath



1.6.4 Fibre Optic Cable

A fibre optic system has three key components: the light source (usually an LED or laser diode), the transmission medium (an ultra-thin glass fibre), and the detector (a photodiode that converts light pulses back into electrical signals).

Light travels down the fibre core through total internal reflection, a phenomenon where light striking the core-to-cladding boundary at a shallow angle bounces back into the core instead of escaping, allowing the signal to travel long distances with very little loss.

Two categories of fibre exist. Multimode fibre has a relatively thick core allowing several light rays (modes) to travel at slightly different angles, useful for shorter distances. Single-mode fibre has a very thin core forcing light to travel in almost a straight line, allowing much higher speed and distance.

Fibre optic cable offers far greater bandwidth, lower attenuation (signal loss over distance), and complete immunity to electrical interference compared to copper-based media, since it uses light rather than electric current.

Fig 1.11 — Fibre optic cable structure showing core, cladding, and light path via total internal reflection

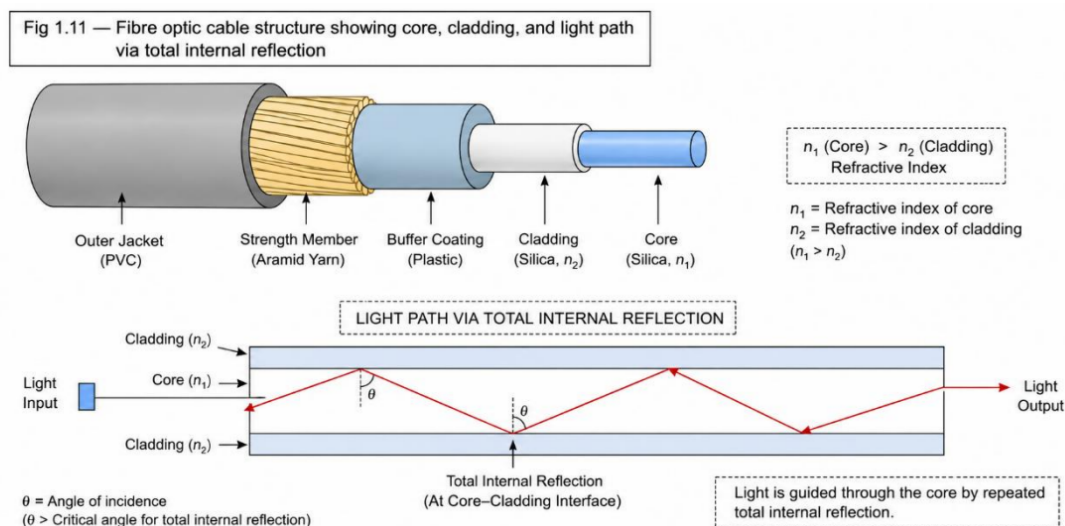


Table: Comparison of Guided Transmission Media

Sr. No.	Parameter	Twisted Pair	Coaxial Cable	Fibre Optic Cable
1	Bandwidth	Low to moderate (up to ~1 Gbps)	Moderate to high (up to ~500 Mbps analog)	Very high (tens of Gbps and beyond)
2	Noise Immunity	Low, prone to crosstalk	Better, due to shielding	Excellent, immune to electrical noise
3	Attenuation	High over distance	Moderate	Very low, supports long distances
4	Cost	Cheapest	Moderate	Most expensive, but falling
5	Typical Use	Telephone lines, office LAN	Cable TV, older Ethernet backbone	Internet backbone, high-speed LAN

Worked Example:

Given: An organisation needs to lay a backbone cable across 40 km between two campuses with zero electrical interference tolerance and very high speed requirement.

Find: Which guided transmission medium should be selected, and why.

Solution:

Step 1: List candidate media: twisted pair, coaxial cable, fibre optic cable.

Step 2: Check distance requirement of 40 km — twisted pair and coaxial cable suffer heavy attenuation at this range.

Step 3: Check interference requirement — fibre optic is immune to electrical interference since it uses light, not current.

Step 4: Check speed requirement — fibre optic supports far higher bandwidth than copper-based options.

Ans: Fibre optic cable should be selected, since it satisfies the long-distance, interference-free, and high-speed requirements.

Advantages of Guided Media (overall):

- Guided media give predictable, stable performance since the path is physically fixed
- Twisted pair is very low cost and easy to install for short-distance LANs
- Coaxial cable gives a good balance of shielding and cost for medium-distance backbones
- Fibre optic offers unmatched bandwidth, distance, and security against tapping
- Physical media generally offer more consistent latency compared to wireless links

Disadvantages / Limitations:

- Physical cables are expensive and time-consuming to install across large areas
- Twisted pair and coaxial cable are vulnerable to electromagnetic interference and tapping
- Fibre optic requires precise alignment and specialised, costly connectors and splicing tools
- Cabling cannot support mobility, unlike wireless transmission

Applications:

- Category 5e/6 twisted pair cabling used in SPPU college computer labs for Ethernet LAN
- RG-6 coaxial cable used by cable operators such as Hathway and Den Networks for television and broadband
- Submarine fibre optic cables such as SEA-ME-WE carrying international Internet traffic across oceans
- Fibre-to-the-home (FTTH) services offered by Jio Fiber and Airtel Xstream for gigabit home Internet
- Fibre optic backbones connecting core routers inside large data centres like those of Google and AWS

KEY POINTS — Guided Transmission Media

Order of bandwidth (low to high): Twisted Pair < Coaxial Cable < Fibre Optic Cable.

Twisting in twisted-pair cable exists purely to cancel crosstalk, not to increase strength.

Common exam trap: students think coaxial cable is obsolete — it is still used heavily in cable TV networks.

Fibre works on total internal reflection — always mention this term explicitly for full marks.

Single-mode fibre gives higher speed and distance than multimode fibre — a frequently confused pair.

SPPU EXPECTED QUESTIONS — Guided Transmission Media

- Q1. Explain the construction and working of twisted pair cable. (5 marks)
- Q2. Describe coaxial cable construction with its two main categories. (5 marks)
- Q3. Explain fibre optic transmission with total internal reflection concept and a neat diagram. (8 marks)
- Q4. Compare twisted pair, coaxial cable, and fibre optic cable. (8 marks)
- Q5. Justify why fibre optic cable is preferred for backbone networks over copper media. (6 marks)

1.7 Wireless Transmission

Definition:

Wireless transmission carries information through free space using electromagnetic waves, without any physical conductor connecting the sender and the receiver.

Introduction / Need:

Guided media cannot serve mobile users, remote areas without cabling infrastructure, or situations where laying cables is too costly or physically impossible, such as across mountains or oceans.

Wireless transmission solves this by using different portions of the electromagnetic spectrum, each suited to a particular range, obstacle tolerance, and data rate.

1.7.1 The Electromagnetic Spectrum

Electromagnetic waves can be made to travel through free space by modulating (varying) their frequency, amplitude, or phase, and the frequency chosen decides the wave's behaviour — low frequencies travel further and bend around obstacles better, while high frequencies travel in straighter lines and carry more data.

The usable spectrum for communication ranges from radio waves through microwaves, infrared, and up to visible light, each occupying a specific band of frequencies regulated by government bodies to avoid interference between users.

1.7.2 Radio Transmission

Radio waves are easy to generate, travel long distances, and pass through obstacles such as walls easily, which is why they are widely used for both indoor and outdoor communication.

Radio waves are omnidirectional, meaning the transmitter does not need to be physically aligned with the receiver, unlike some higher-frequency systems that require line of sight.

At low frequencies, radio waves pass through obstacles well but the signal power falls sharply with distance. At high frequencies, radio waves tend to travel in straight lines and can be reflected off obstacles, causing multipath interference.

1.7.3 Microwave Transmission

Above 100 MHz, waves travel in nearly straight lines and can be tightly focused, so microwave transmission requires the sending and receiving antennas to be precisely aligned, a requirement called line-of-sight communication.

Since the waves travel in a straight line and the earth is curved, tall towers are needed to extend range; microwave repeater towers are typically installed every few tens of kilometres to relay the signal further.

Microwave transmission is heavily used for long-distance telephone and television communication, and also for point-to-point links between buildings where laying cable is impractical.

1.7.4 Infrared and Millimetre Waves

Infrared and millimetre waves are used for short-range wireless communication, such as television remote controls and short indoor device links, since they are relatively directional, cheap, and easy to build.

A major limitation is that infrared waves do not pass through solid objects such as walls, which actually helps prevent interference between devices in adjacent rooms but limits range to line-of-sight within one room.

1.7.5 Lightwave (Laser) Transmission

Coherent, single-frequency laser light can be transmitted through open air to link networks in different buildings, using a laser transmitter on one rooftop aimed at a photodetector on another.

Laser links are cheap, need no government licensing, and offer very high bandwidth, but suffer badly from weather conditions such as fog, heavy rain, and temperature-based air turbulence that can bend or scatter the beam.

Fig 1.12 — Electromagnetic spectrum showing radio, microwave, infrared, and light regions used for wireless transmission

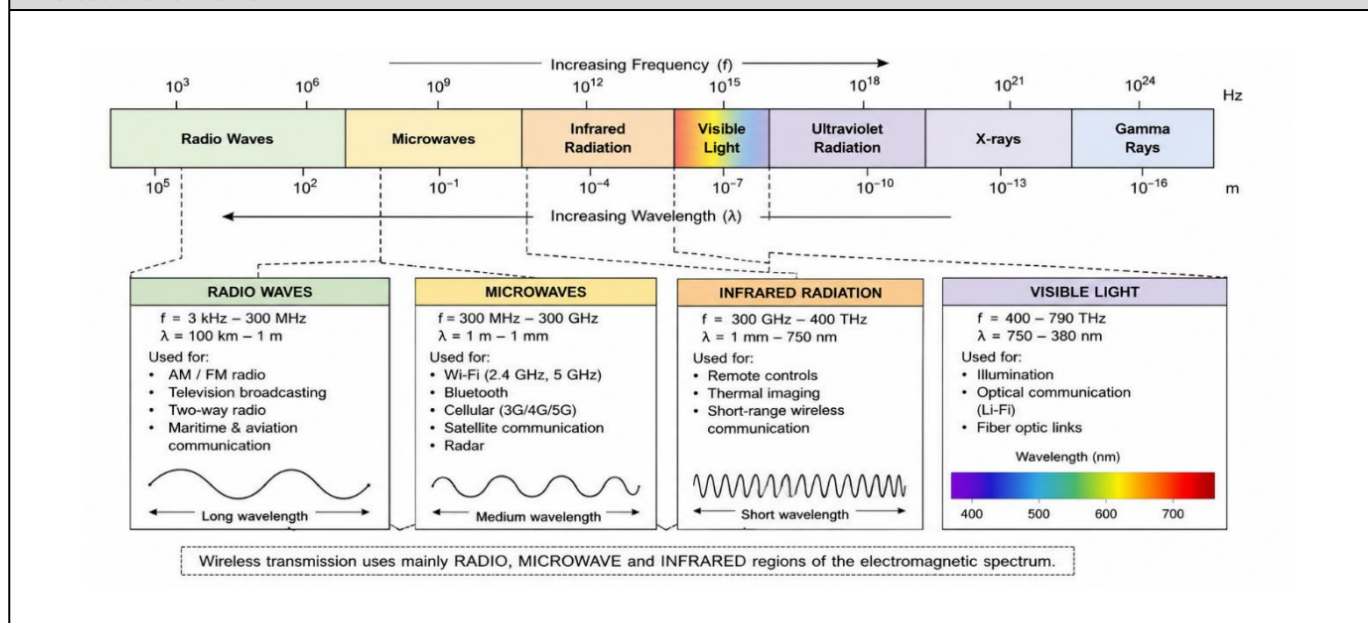


Table: Comparison of Wireless Transmission Types

Sr. No.	Parameter	Radio Waves	Microwaves	Infrared / Laser
1	Directionality	Omnidirectional	Line-of-sight, directional	Highly directional
2	Obstacle Penetration	Good, passes through walls	Poor, blocked by obstacles	Blocked completely by walls
3	Range	Long range	Long range with repeater towers	Short range (same room/building)
4	Typical Use	Wi-Fi, radio broadcast	Long-distance telephone, satellite links	TV remote, rooftop laser LAN links
5	Interference Tolerance	Moderate	Low, needs precise alignment	Low, affected by weather/fog for laser

Worked Example:

Given: Two office buildings 500 metres apart, separated by a public road, need a quick wireless link without government licensing and with very high bandwidth.

Find: Suggest a suitable wireless transmission technology.

Solution:

Step 1: Eliminate radio waves, since regulated frequencies mostly need licensing for long-range dedicated links.

Step 2: Eliminate microwave towers, since the short distance does not justify tower infrastructure cost.

Step 3: Consider laser/lightwave transmission, which needs no licensing and gives very high bandwidth.

Step 4: Confirm line-of-sight is possible between rooftops of both buildings across the 500 m gap.

Ans: A rooftop laser (lightwave) link is the most suitable choice for this short, unlicensed, high-bandwidth requirement.

Advantages of Wireless Transmission:

- Enables mobility — users can connect while moving, unlike guided media
- Faster deployment in remote or difficult terrain where cabling is impractical
- Radio waves can penetrate walls, useful for indoor coverage without extra cabling
- Laser and infrared links need no government frequency licensing

Disadvantages / Limitations:

- Susceptible to interference from other signals and physical obstacles
- Microwave and laser links require precise line-of-sight alignment
- Weather conditions such as rain and fog can degrade laser and microwave signal quality
- Generally lower security compared to guided media unless encrypted

Applications:

- Wi-Fi routers in homes and colleges using radio-frequency wireless transmission
- Satellite television and long-distance telephone trunks using microwave transmission
- TV and air-conditioner remote controls using infrared transmission
- Rooftop laser links used by ISPs for last-mile connectivity between nearby buildings

KEY POINTS — Wireless Transmission

Lower frequency = better obstacle penetration; higher frequency = higher data rate but line-of-sight needed.

Microwave transmission strictly needs line-of-sight alignment — a favourite one-line exam fact.

Common exam trap: students confuse infrared with microwave — infrared cannot pass through walls at all.

Laser transmission needs no licensing but is heavily weather-dependent — always mention both sides.

SPPU commonly asks to classify a given scenario into radio, microwave, or infrared — practice scenario questions.

SPPU EXPECTED QUESTIONS — Wireless Transmission

Q1. Explain radio transmission with its characteristics. (5 marks)

Q2. Describe microwave transmission and explain why line-of-sight is required. (6 marks)

Q3. Compare radio, microwave, and infrared transmission. (8 marks)

Q4. Explain lightwave (laser) transmission with its advantages and limitations. (5 marks)

1.8 The Telephone System

Definition:

The telephone system is a globally interconnected, circuit-switched communication network originally designed for voice calls, whose infrastructure is also widely reused today for computer data transmission.

Introduction / Need:

Long before computer networks existed, the telephone system had already built a massive, reliable, worldwide wiring infrastructure connecting almost every home and office.

Understanding the telephone system matters because early wide area computer networks, and even many broadband technologies today, were built by reusing this existing telephone infrastructure rather than laying entirely new cables.

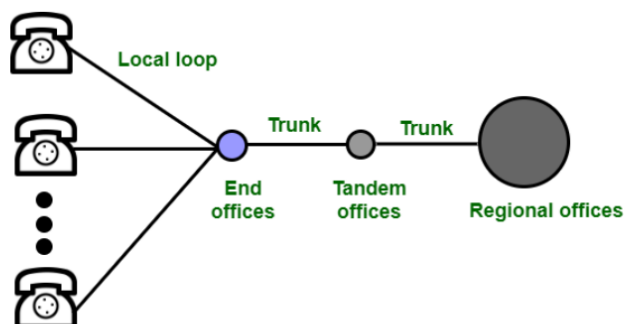
1.8.1 Structure of the Telephone System

The telephone system has three major components: local loops (analog twisted pair wires running from each subscriber's phone to the nearest telephone exchange), trunks (high-capacity digital fibre or microwave links connecting exchanges together), and switching offices (equipment that connects one line to another to complete a call).

A call from one home phone to another typically passes through: the local loop to the nearest end office (also called a local exchange), then possibly through one or more toll offices and trunk lines, and finally down another local loop to reach the destination phone.

Because the local loop was originally designed only for voice-range analog signals (roughly 300 Hz to 3400 Hz), a modem (modulator-demodulator, a device converting digital computer signals into analog signals and back) was historically required to send computer data over it.

Fig 1.13 — Structure of telephone network showing local loop, trunks, and switching offices



1.8.2 Politics of the Telephone System

Telephone systems worldwide were historically run either as government-owned monopolies (as in most of Europe and Asia) or as regulated private monopolies (as with AT&T in the United States before its breakup).

Deregulation over the past decades has opened telephone infrastructure to competition, allowing multiple private telecom operators to offer service using shared or leased infrastructure, which is the model followed by most countries including India today.

1.8.3 Evolution Towards Digital Transmission

Modern telephone trunks between exchanges have been almost fully converted to digital transmission using fibre optic cable, while the last stretch (local loop) to homes remains largely analog copper wire in many older installations.

Technologies such as Digital Subscriber Line (DSL) were developed specifically to push higher digital bandwidth over these existing analog copper local loops without replacing the wiring.

Table: Components of the Telephone System

Sr. No.	Component	Function
1	Local Loop	Analog twisted pair wire connecting a subscriber's phone to the nearest exchange
2	Trunk	High-capacity digital fibre/microwave link connecting exchanges over long distance
3	Switching Office	Equipment that connects an incoming call line to the correct outgoing line
4	Modem	Converts digital computer signal to analog for transmission over the analog local loop, and back

Worked Example:

Given: A user's home computer must send digital data over an old analog telephone local loop to reach the Internet Service Provider.

Find: Identify what device is required, and why.

Solution:

Step 1: Recall that the local loop was originally engineered only for analog voice signals.

Step 2: A computer produces digital signals which cannot travel correctly over an analog-only line.

Step 3: A modem is required to convert (modulate) digital signal into analog form for transmission.

Step 4: At the receiving end, another modem demodulates the analog signal back into digital form.

Ans: A modem is required at both ends to convert between digital computer data and analog telephone line signals.

Advantages of the Telephone System (as data carrier):

- Already existing, widespread infrastructure reaching almost every household
- Trunk lines are now digital and fibre-based, offering high reliability and speed
- Well-established switching and billing systems, mature over a century of use
- Backbone of technologies such as DSL broadband still used in many regions

Disadvantages / Limitations:

- Local loop analog design limits maximum achievable bandwidth compared to fibre-to-home
- Legacy copper wiring in some areas is old, degraded, and prone to noise
- Historically monopolistic structure slowed early competition and innovation

Applications:

- DSL broadband services (such as BSNL landline broadband) reusing existing local loop wiring
- Fax and older dial-up Internet services using modems over the telephone system
- Long-distance voice trunking between exchanges using fibre and microwave links

- Emergency and landline voice services still relying on this century-old infrastructure

KEY POINTS — Telephone System

Three core components to remember: Local Loop, Trunk, Switching Office.

Local loop is analog; modern trunks are digital — this mismatch is why modems were invented.

Common exam trap: students think entire telephone system is analog — only the local loop mostly is.

DSL was built specifically to exploit the existing analog local loop for higher bandwidth.

SPPU often asks to draw and label the local loop-trunk-switching office structure.

SPPU EXPECTED QUESTIONS — Telephone System

Q1. Explain the structure of the telephone system with a neat diagram. (6 marks)

Q2. What is the role of a modem in telephone-based data transmission? (5 marks)

Q3. Describe local loop, trunk, and switching office in the telephone network. (6 marks)

1.9 Narrowband and Broadband Communication Systems

Definition:

Narrowband and broadband communication systems classify data transmission based on the range of frequencies (bandwidth) a channel can carry, and consequently, how much data it can transmit per second.

Introduction / Need:

Not every communication requirement needs the same channel capacity. A simple sensor sending occasional temperature readings needs very little bandwidth, while a 4K video stream needs enormous bandwidth.

Classifying systems as narrowband or broadband helps engineers and service providers design and price network connections appropriately for the intended use case.

1.9.1 Narrowband Communication

Narrowband systems use a relatively small range of frequencies (a narrow band), giving low data rates but excellent range and obstacle penetration, historically used for basic telephone voice channels and early low-speed modems.

Because the frequency band is narrow, multiple narrowband channels can be packed efficiently into a limited spectrum, which is useful in applications such as walkie-talkies, AM radio, and legacy dial-up modems.

1.9.2 Broadband Communication

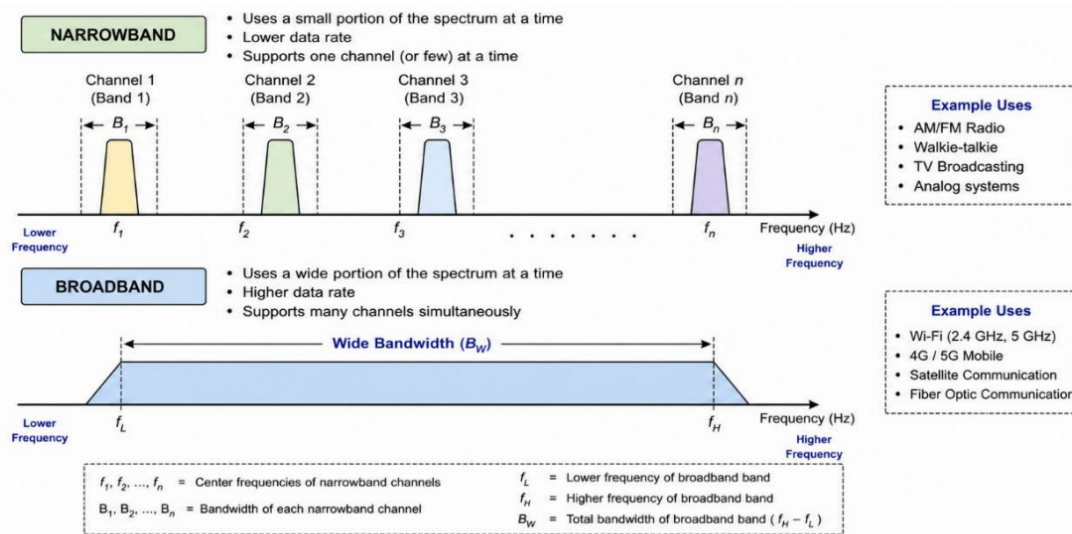
Broadband systems use a much wider range of frequencies, allowing many independent channels or a very high data rate to be carried simultaneously on the same physical medium, using techniques like frequency-division multiplexing.

Broadband is the basis of modern high-speed Internet access technologies such as DSL, cable modem, and fibre optic connections, which can carry voice, data, and video simultaneously on the same line.

Sr. No.	Parameter	Narrowband	Broadband
1	Bandwidth	Narrow frequency range	Wide frequency range

Sr. No.	Parameter	Narrowband	Broadband
2	Data Rate	Low (few kbps typically)	High (Mbps to Gbps)
3	Range and Penetration	Longer range, better obstacle penetration	Shorter effective range at high frequency without repeaters
4	Example Technology	AM radio, dial-up modem, walkie-talkie	DSL, cable Internet, fibre optic broadband
5	Typical Use Case	Basic voice, low-rate telemetry/IoT sensors	Video streaming, high-speed Internet, multimedia

Fig 1.14 — Narrowband vs broadband frequency spectrum usage comparison



Worked Example:

Given: A smart agricultural sensor needs to send a small temperature reading once every hour across several kilometres with minimal power usage.

Find: Should the system use narrowband or broadband communication, and why.

Solution:

Step 1: Identify data volume: a single small reading per hour is very low data volume.

Step 2: Identify range requirement: several kilometres, favouring longer-range transmission.

Step 3: Recall that narrowband systems achieve longer range and lower power draw for small data volumes.

Step 4: Broadband would be unnecessary and power-inefficient for this low-data, long-range use case.

Ans: Narrowband communication is the correct choice, since it suits low data volume, long range, and low power needs.

Applications:

- Narrowband IoT (NB-IoT) used in smart water and gas meters for long-range, low-power reporting
- AM radio broadcasting using narrowband frequency allocation for long-distance audio
- Broadband fibre-to-home services such as Jio Fiber for streaming and multi-device households
- Cable broadband Internet from operators such as Hathway carrying both TV and Internet on one line

KEY POINTS — Narrowband and Broadband Systems

Narrowband = low bandwidth, long range, low power; Broadband = high bandwidth, high data rate.

Common exam trap: students assume broadband always means "Internet" — broadband is a bandwidth classification, not a specific service.

NB-IoT is a modern, frequently asked real-world example of narrowband communication.

Broadband commonly uses frequency-division multiplexing to carry multiple channels together.

SPPU often asks to classify a given real-world scenario as narrowband or broadband.

SPPU EXPECTED QUESTIONS — Narrowband and Broadband Systems

Q1. Differentiate between narrowband and broadband communication systems. (6 marks)

Q2. Explain broadband communication with real-world examples. (5 marks)

Q3. Justify the choice of narrowband communication for IoT sensor networks. (6 marks)

Unit I Summary**UNIT I SUMMARY**

1.1 Introduction — A network is autonomous computers connected using common protocols; differs from distributed systems.

1.2 Uses — Networks serve business (resource/data sharing), home (entertainment/IoT), and mobile (location-based) needs.

1.3 Network Hardware — Classification by scale: PAN < LAN < MAN < WAN < Internetwork; WAN has hosts + subnet with routers.

1.4 Network Software — Layered protocol hierarchy; services (vertical) vs protocols (horizontal); connection-oriented vs connectionless.

1.5 Reference Models — OSI has 7 layers (teaching model); TCP/IP has 4 layers (actual Internet model).

1.6 Guided Media — Twisted pair (cheap, low BW), coaxial (shielded, medium BW), fibre optic (light-based, highest BW, uses total internal reflection).

1.7 Wireless Transmission — Radio (omnidirectional, long range), microwave (line-of-sight, towers), infrared/laser (short range, no wall penetration).

1.8 Telephone System — Local loop (analog) + trunk (digital) + switching office; modem bridges digital data over analog loop.

1.9 Narrowband/Broadband — Narrowband: low data rate, long range, low power (NB-IoT); Broadband: high data rate (DSL, fibre, cable).

Important formula/relationship: Service defines WHAT a layer offers upward; Protocol defines HOW peer layers communicate — most repeated exam distinction.

Key comparison asked every year: OSI vs TCP/IP layer mapping, and LAN vs MAN vs WAN.

Most common SPPU exam question this unit: "Explain OSI and TCP/IP reference models and compare them" — always prepare this in full detail.